Student AI Usage & Academic Performance Analysis

GitHub Project Report

1. Project Overview

This project analyzes 7,200 student records to understand how Artificial Intelligence (AI) usage, screen time, social media usage, study habits, sleep, physical activity, wellbeing, and academic performance relate to one another. The project follows an end-to-end Data Analyst workflow using Excel, Python, SQL, and Power BI.

2. Project Objectives

- Analyze student demographics and academic performance.
- Understand AI usage patterns and identify commonly used AI tools.
- Analyze the purposes for which students use AI.
- Analyze screen time and social media usage.
- Compare AI usage with GPA and screen time with GPA.
- Analyze study hours, sleep patterns, focus, and academic performance.
- Analyze wellbeing indicators such as anxiety, attention, and physical activity.
- Analyze parental control patterns.
- Build an interactive Power BI dashboard for reporting and exploration.

3. Dataset

The dataset contains 7,200 student records with demographic, AI usage, digital behavior, academic, lifestyle, and wellbeing variables.

Category	Examples
Student Information	Student ID, Age, Age Group, Gender, City
AI Usage	Daily AI Usage Hours, AI Usage Level, Primary AI Tool, AI Purpose
Digital Usage	Screen Time, Screen Time Level, Social Media Hours, Social Media Level
Academic	Study Hours, Study Level, GPA, GPA Level
Lifestyle	Sleep Hours, Sleep Level, Physical Activity, Physical Activity Level
Wellbeing	Attention, Focus, Anxiety, Anxiety Level, Parental Control

4. Tools & Technologies

Tool	Purpose
Microsoft Excel	Data cleaning and preparation
SQL	Data querying and analysis
Python / Pandas	Data cleaning and exploratory analysis

Power BI	Interactive dashboards and visualization
DAX	Measures and KPI calculations

5. Project Workflow

Dataset
↓
Excel Data Cleaning
↓
Python Data Cleaning & Analysis
↓
SQL Analysis
↓
Power BI Data Modeling
↓
DAX Measures
↓
Interactive Dashboard
↓
Reports & Insights

6. Power BI Dashboard

The Power BI report is organized into four analytical pages so that the overall dashboard remains clear while detailed analysis is available in separate sections.

Page 1 — Student Overview

• KPI Cards: Total Students, Average GPA, Average AI Usage, Average Study Hours, Average Sleep Hours, Average Anxiety.

• Students by City — Clustered Column Chart.

• Students by Gender — Donut Chart.

• Students by Age Group — Clustered Column Chart.

• AI Usage Level — Donut Chart.

• GPA Level — Clustered Column Chart.

• Slicers: City, Gender, Age Group, AI Usage Level, GPA Level.

Page 2 — AI & Digital Usage

• AI Usage by AI Tool — Bar Chart.

• AI Usage by Purpose — Bar Chart.

• Screen Time by Level — Donut Chart.

• Social Media Hours — Column Chart.

• AI Usage vs GPA — Scatter Chart.

• Screen Time vs GPA — Scatter Chart.

Page 3 — Academic Performance

• GPA by Study Level — Clustered Column Chart.

• GPA by Sleep Level — Clustered Column Chart.

• GPA by Focus Level — Clustered Column Chart.

• Study Hours vs GPA — Scatter Chart.

• AI Usage vs GPA — Scatter Chart.

• Screen Time vs GPA — Scatter Chart.

Page 4 — Student Wellbeing

• Anxiety Level — Donut Chart.

• Attention Level — Donut Chart.

• Physical Activity Level — Donut Chart.

• Parental Control — Donut Chart.

• GPA by Sleep Level — Column Chart.

• GPA by Focus Level — Column Chart.

7. Key DAX Measures

Total Students

Total Students = DISTINCTCOUNT(Sheet1[Student_ID])

Average GPA

Average GPA = AVERAGE(Sheet1[GPA_10])

Average AI Usage

Average AI Usage = AVERAGE(Sheet1[Daily_AI_Usage_Hrs])

Average Study Hours

Average Study Hours = AVERAGE(Sheet1[Daily_Study_Hrs])

Average Sleep Hours

Average Sleep Hours = AVERAGE(Sheet1[Daily_Sleep_Hrs])

Average Anxiety

Average Anxiety = AVERAGE(Sheet1[Anxiety_Score_1_10])

8. Data Cleaning & Analysis

• Checked missing values and handled them where appropriate.

• Checked duplicate records and data consistency.

• Verified numerical and categorical data types.

• Standardized column names and category values.

• Created or used meaningful category levels such as Age Group, AI Usage Level, Screen Level, Study Level, Sleep Level, GPA Level, Anxiety Level, Attention Level, Focus Level, and Physical Activity Level.

• Used Python/Pandas for inspection and exploratory analysis and SQL for analytical queries.

9. Analysis Questions

• How many students are included in the dataset?

• How are students distributed by city, gender, and age group?

• Which AI tools are most commonly used?

• What are the main purposes of AI usage?

• How is screen time distributed across students?

• How much social media time do students report?

• How does GPA vary by study, sleep, and focus levels?

• What patterns can be observed between AI usage and GPA?

• What patterns can be observed between screen time and GPA?

• What patterns can be observed between study hours and GPA?

• How are anxiety, attention, physical activity, and parental control distributed?

10. Interpretation

Scatter charts and grouped comparisons are used to identify patterns and associations in the data. For example, AI Usage vs GPA, Screen Time vs GPA, and Study Hours vs GPA can help identify whether variables move together. These relationships should be interpreted as associations rather than proof of causation; additional statistical analysis would be needed to establish causal relationships.

11. GitHub Repository Structure

```
student-ai-academic-performance/
■
■■■ README.md
■■■ PROJECT_REPORT.md
■■■ data/
■   ■■■ student_dataset.xlsx
■■■ python/
■   ■■■ data_analysis.py
■■■ sql/
■   ■■■ analysis_queries.sql
■■■ powerbi/
■   ■■■ student_dashboard.pbix
■■■ screenshots/
■   ■■■ overview_dashboard.png
■   ■■■ ai_digital_usage.png
■   ■■■ academic_performance.png
■   ■■■ student_wellbeing.png
■■■ documentation/
    ■■■ project_notes.md
```

12. Skills Demonstrated

Excel • Data Cleaning • SQL • Python • Pandas • Exploratory Data Analysis • Power BI • DAX • Data Visualization • KPI Development • Dashboard Design • Interactive Reporting • Data Storytelling

13. Conclusion

This project demonstrates an end-to-end Data Analyst workflow, from data cleaning and preparation through SQL and Python analysis to interactive Power BI reporting. The final dashboard provides a structured view of student demographics, AI usage, digital behavior, academic performance, and wellbeing, allowing users to explore patterns and relationships through interactive filters and visuals.